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April 8, 2026

Emilio Ferrer, Ph.D.
Editor-in-Chief, *Multivariate Behavioral Research*

Dear Dr. Ferrer:

We are pleased to submit our manuscript, **From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models**, for publication in *Multivariate Behavioral Research*.

This manuscript introduces *MetaVAR*, a two-stage meta-analytic structural equation modeling (MASEM) framework for drawing population-level inferences from person-specific multilevel dynamic models. In Stage 1, person-specific vector autoregressive (VAR) models are fit separately for each individual to obtain a dynamic parameter vector and its sampling covariance matrix. In Stage 2, these estimates are synthesized using a multivariate random-effects meta-analytic SEM. This framework supports inference on population-average dynamics, between-person heterogeneity, and cross-parameter covariation while explicitly propagating first-stage uncertainty.

We believe the manuscript is well suited for *Multivariate Behavioral Research* because it addresses a central methodological problem in intensive longitudinal research: how to move from idiographic estimation to valid multivariate population-level inference without collapsing the distinction between within-person and between-person structure. More broadly, the paper integrates intensive longitudinal modeling, multilevel dynamic modeling, multivariate meta-analysis, and meta-analytic structural equation modeling into a practical framework for complex multivariate longitudinal data.

The manuscript makes three main contributions. First, it formalizes a general two-stage framework in which persons are treated as the units of synthesis and person-specific dynamic signatures are modeled as multivariate effect-size vectors. Second, a Monte Carlo simulation study shows that MetaVAR's clearest advantage lies in recovering between-person heterogeneity and improving interval calibration for heterogeneity parameters relative to both an uncertainty-uncorrected two-stage benchmark and a one-step Bayesian multilevel VAR benchmark. Third, an empirical illustration using ecological momentary assessment data on negative and positive affect shows how MetaVAR can recover interpretable average dynamics together with substantial heterogeneity in affective set points, inertia, cross-lagged coupling, and innovation structure.

We believe this manuscript will be of interest to readers of *Multivariate Behavioral Research* who develop, evaluate, or apply multivariate methods for longitudinal, intensive longitudinal, and dynamic data. The manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved the final version and consent to its submission.

Thank you for your time and consideration.

Sincerely,

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